

SC2000/CZ2100

Probability & Statistics

Week 5-6

Prob. Distribution of Continuous r.v.

- Introduction**
- Special Probability Distributions**
 - Uniform Distribution
 - Exponential Distribution
 - Normal Distribution

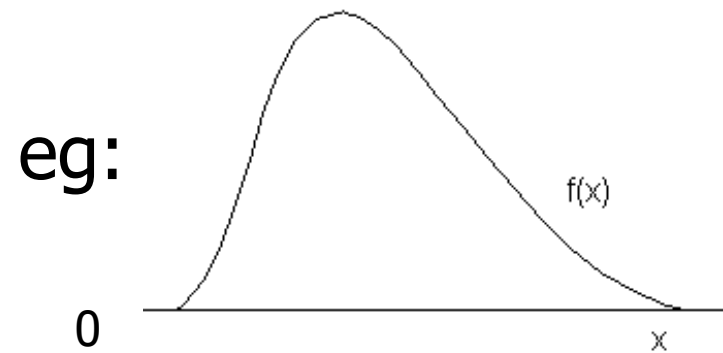
Prob. Distribution of Continuous r.v.

A continuous random variable is a variable used for continuous measurements such as distance, weight, and time.

Probability Density Function (pdf)

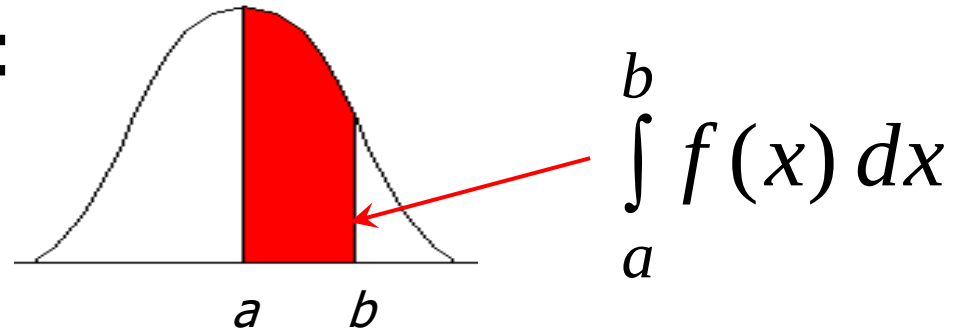
For a continuous random variable X , we can describe the probability distribution by some function $f(X)$, such that

(i) $f(X) \geq 0$ for all X



Prob. Distribution of Continuous r.v.

(ii) $P(a < X < b)$ is the area under the curve between a and b :



(iii) Total area under curve = 1.

The function $f(x)$ is called the probability density function (or pdf) of X .

The probability $P(X < x)$ is known as the Cumulative distribution function (or CDF) of X .

Given that the probability density function of X is:

$$f(x) = \frac{x^3}{4} \text{ where } 0 < x < c$$

Find the value of the constant c.

$$\int_0^c \frac{x^3}{4} dx = \frac{x^4}{16} \Big|_0^c = 1 \quad \text{Solve for } c, \text{ we get } c = 2.$$

Determine the CDF of X.

$$\begin{aligned} \text{CDF of X} &= \int_{-\infty}^x f(x) dx \\ &= \int_{-\infty}^x \frac{x^3}{4} dx = \frac{x^4}{16} \Big|_0^x = \frac{x^4}{16} \end{aligned}$$

Prob. Distribution of Continuous r.v.

For a continuous random variable X the probability that X equals to an exact value is zero.

Eg: for the length of a bolt

$$P(X = 1.999965700 \text{ cm}) = 0$$

i.e., $P(X=x) = 0$. Hence, for continuous random variables, probabilities are always associated with a range of values.

Eg: $P(X \leq 2.00 \text{ cm})$ or $P(X > 1.95 \text{ cm})$.

Prob. Distribution of Continuous r.v.

The expected value of a continuous r.v. is defined

$$\text{as: } E[X] = \int_{x=-\infty}^{x=\infty} x f(x) dx = \mu$$

For discrete r.v.

$$E(X) = \sum_{i=1}^n p_i x_i$$

and its variance is:

$$\begin{aligned} \text{var}[X] &= \int_{x=-\infty}^{x=\infty} (x - \mu)^2 f(x) dx \\ &= E[X^2] - E[X]^2 \end{aligned}$$

For discrete r.v.

$$\text{Var}[X] = \sum_{i=1}^n (x_i - \mu)^2 p_i$$

$$\text{where } E[X^2] = \int_{x=-\infty}^{x=\infty} x^2 f(x) dx$$

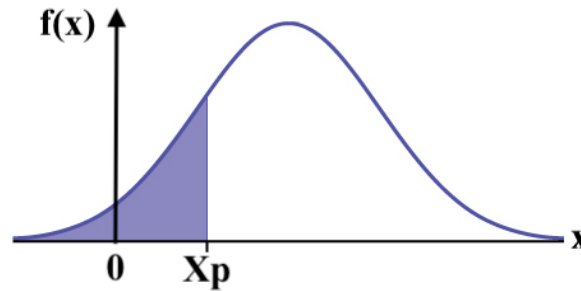
Example:

The pdf of X is given below. Determine $E[X]$.

$$f(x) = \begin{cases} \cos(x), & 0 \leq x \leq \frac{\pi}{2} \\ 0, & \text{otherwise} \end{cases}$$

Prob. Distribution of Continuous r.v.

How do we determine the p^{th} percentile from the probability distribution of a continuous r.v.?



The p^{th} percentile is the value X_p such that the probability that $x < X_p$ is p percent.

$$\int_{-\infty}^{X_p} f(x) dx = \frac{p}{100}$$

$$p^{\text{th}} \text{ percentile} = X_p$$

Given that the probability density function of x is

$$f(x) = e^{-x}, \text{ where } x > 0$$

determine the 25th percentile.

Prob. Distribution of Continuous r.v.

- Introduction

- **Special Probability Distributions**

- Uniform Distribution
- Exponential Distribution
- Normal Distribution

Prob. Distribution of Continuous r.v.

Uniform Distribution

One example of a pdf for a continuous r.v. is the uniform continuous distribution.

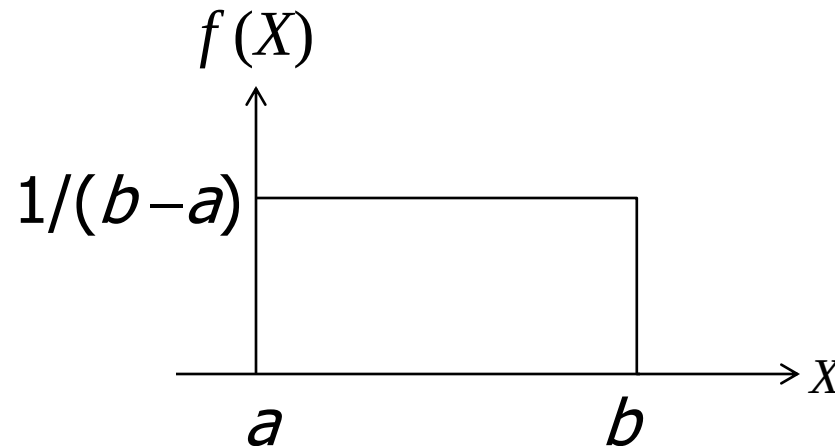
X can take any real value between a and b with uniform probability over this interval.

Notation: $X \sim U(a, b)$.

Total area = 1

Length = $b - a$

Height = $1/(b - a)$

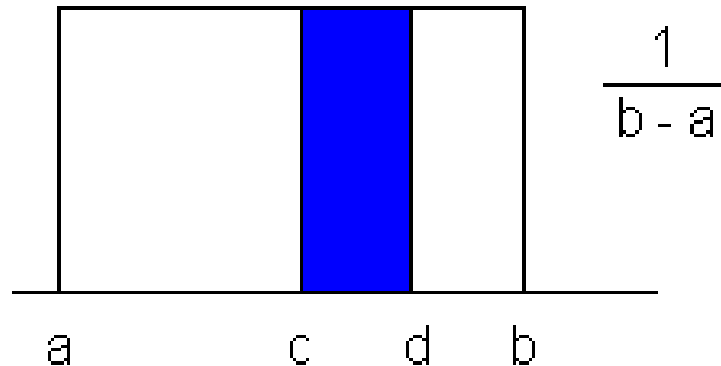


Prob. Distribution of Continuous r.v.

Thus the pdf is $f(X) = \frac{1}{b-a}$ for $a \leq X \leq b$
 $= 0$, otherwise

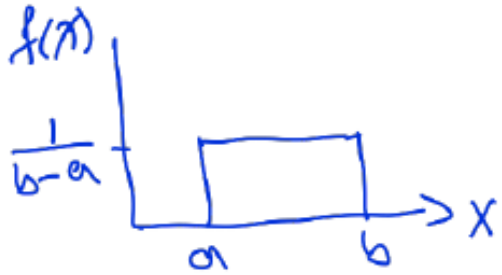
For any values c and d between a and b

$$P(c < X < d) = \frac{1}{b-a} (d - c)$$



Exercise: Obtain the CDF.

Obtain the expressions for $E[X]$ and $\text{Var}[X]$ of r.v. $X \sim U(a,b)$.



$$\begin{aligned} E[X] &= \int_a^b x f(x) dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b \\ &= \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{(b-a) 2} \end{aligned}$$

$$\begin{aligned} \text{Var}[X] &= E[X^2] - E[X]^2 \\ &= \frac{1}{b-a} \int_a^b x^2 dx \\ &= \frac{b^3 - a^3}{3(b-a)} = \frac{(a^2 + ab + b^2)(b-a)}{3(b-a)} \end{aligned}$$

$$\begin{aligned} \therefore \text{Var}[X] &= \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} \\ &= \frac{a^2 + b^2 - 2ab}{12} = \frac{(a-b)^2}{12} \end{aligned}$$

Prob. Distribution of Continuous r.v.

Exponential Distribution

If the number of arrivals during an interval is Poisson distributed, then the interarrival times x are exponentially distributed. The pdf of x is:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

where λ is the mean arrival rate and the mean interarrival time is $1/\lambda$.

Notation: $X \sim \text{Exp}(\lambda)$

If the number of arrivals in an interval is Poisson distributed, we show that the interarrival times are exponentially distributed.

Prob. Distribution of Continuous r.v.

Exponential Distribution – examples:

Pdf:

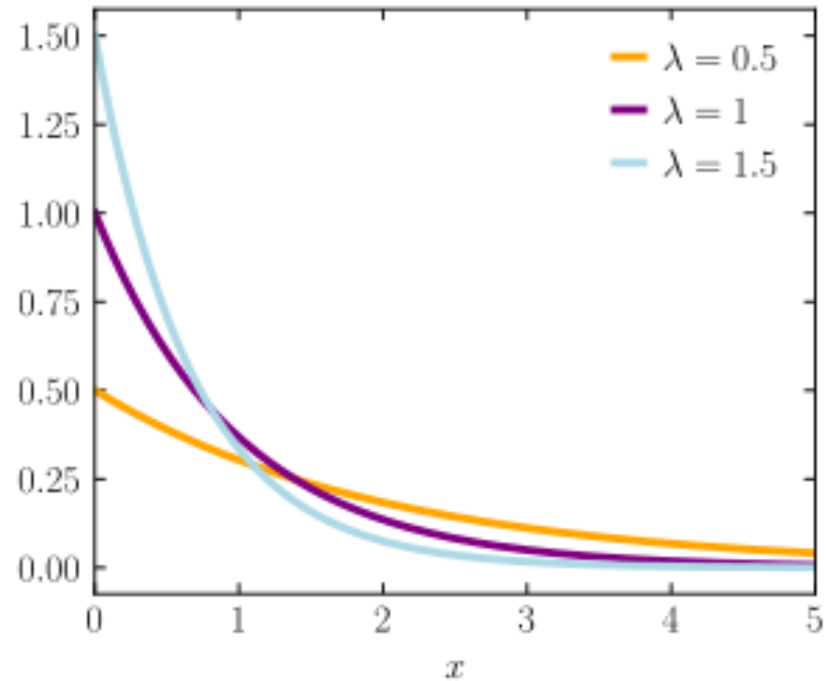
$$f(x) = \lambda e^{-\lambda x}$$

Expected value:

$$E[X] = \frac{1}{\lambda}$$

Variance:

$$\text{Var}[X] = \frac{1}{\lambda^2}$$



Eg: Given that an average of number of customers walk into a shop per hour is 20 and the time between arrivals is exponentially distributed. After a customer arrives, what is the probability that the next customer will arrive within 1 minute?

Prob. Distribution of Continuous r.v.

Normal Distribution

A common function for a continuous distribution is the normal (bell-shaped) curve. Its pdf depends on μ and σ , and is given by:

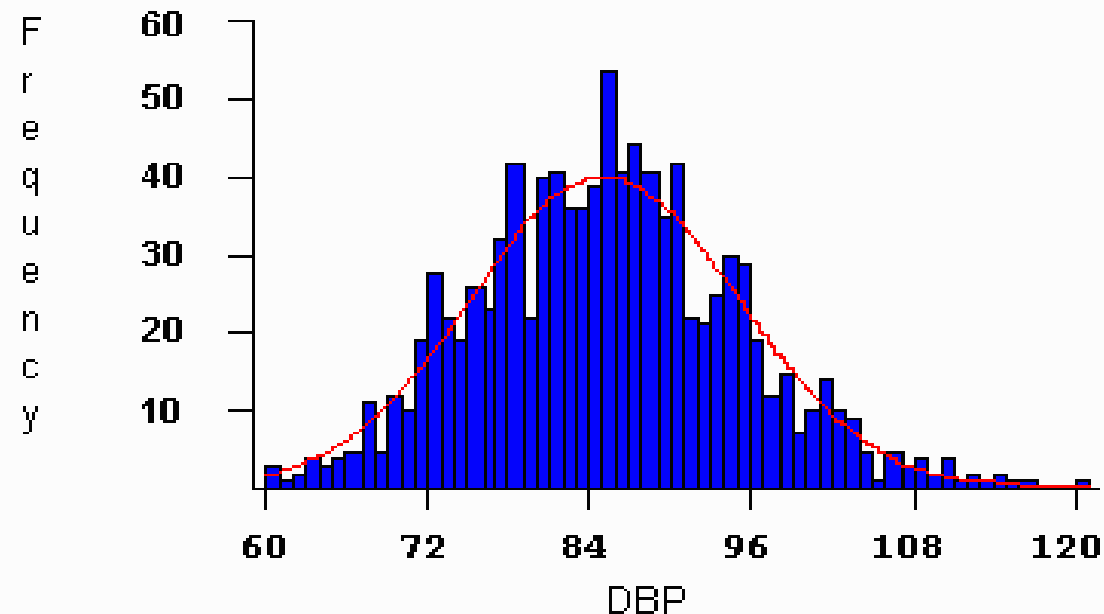
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{where } -\infty < x < \infty$$

The normal distribution is by convention written as $X \sim N(\mu, \sigma^2)$. Eg: if $X \sim N(10, 16)$ then this implies that $\mu=10$ and $\sigma^2=16$, and (hence $\sigma=4$).

Prob. Distribution of Continuous r.v.

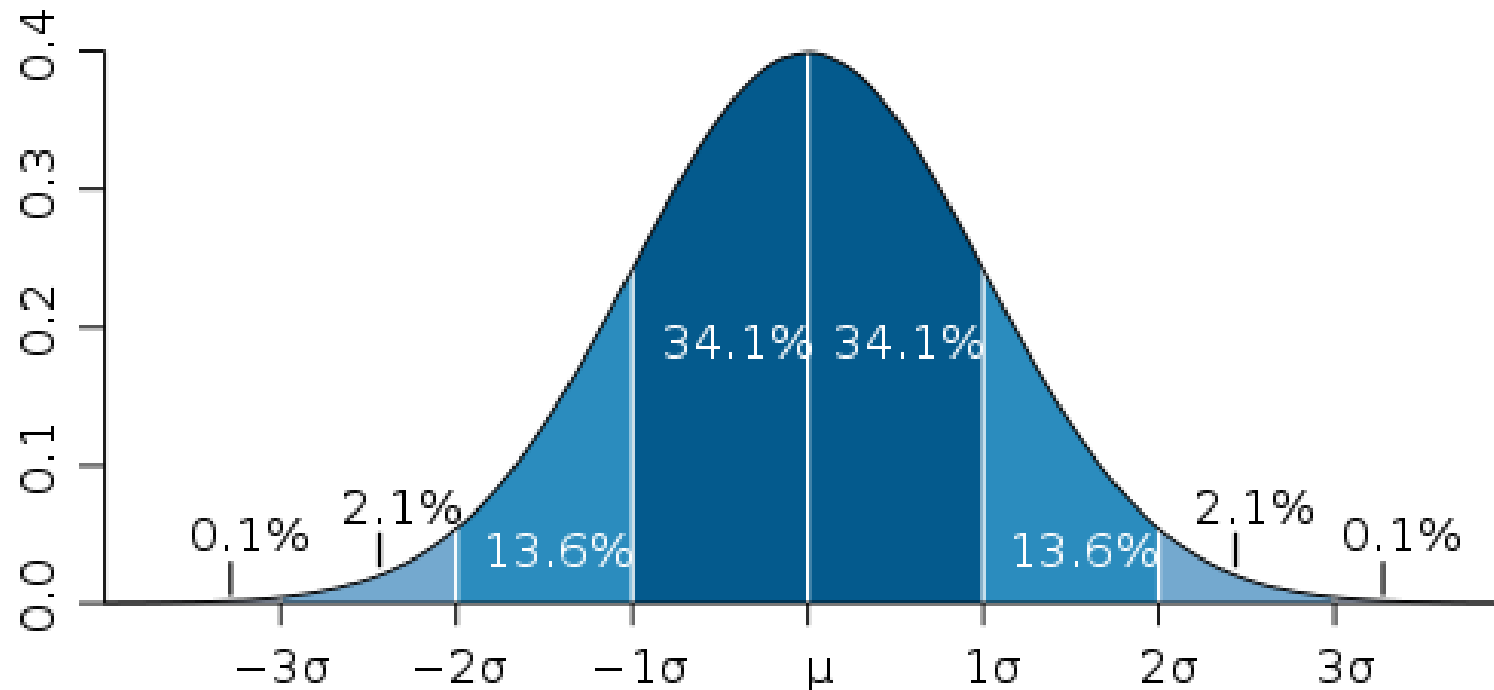
Eg: The distribution of blood pressure in patients can be approximated as a normal distribution with mean 85 mmHg and standard deviation 20 mmHg.

A histogram of 1,000 observations and the normal curve is shown below.



Prob. Distribution of Continuous r.v.

The normal distribution has about 68% of the observations lying within one standard deviation of the mean, 95% within two standard deviations and 99.7% within 3 standard deviations.

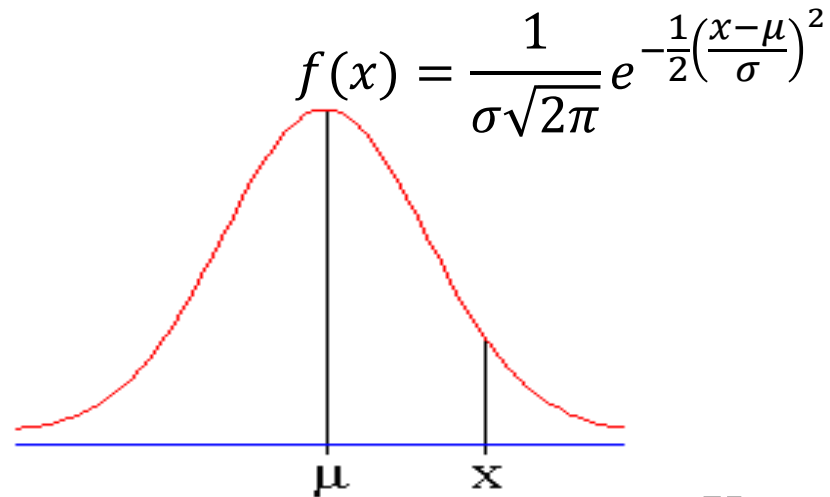


Prob. Distribution of Continuous r.v.

Standard Normal Distribution

To obtain the area under the density curve for a normal distribution, it is necessary to express any value of X in terms of the number of standard deviation units away from μ , i.e.

$$X = \mu + Z\sigma.$$



This can be rearranged to give $Z = \frac{X - \mu}{\sigma}$

Prob. Distribution of Continuous r.v.

Since X is a r.v., so is Z . Using the formulas for functions of random variables we can obtain

$$E[Z] = \frac{1}{\sigma} E[X] - \frac{1}{\sigma} \mu = \frac{1}{\sigma} \mu - \frac{1}{\sigma} \mu = 0$$

$$\text{var}[Z] = \frac{1}{\sigma^2} \text{var}[X] = \frac{\sigma^2}{\sigma^2} = 1$$

Also it can be shown that Z has a normal distribution. Thus $Z \sim N(0, 1)$.

This is called the standard normal distribution.

Prob. Distribution of Continuous r.v.

Most statistics textbooks include a table of the standard Normal distribution. It tabulates the area under the curve of the $N(0,1)$ density function.

Other normal tables may give other areas, e.g.

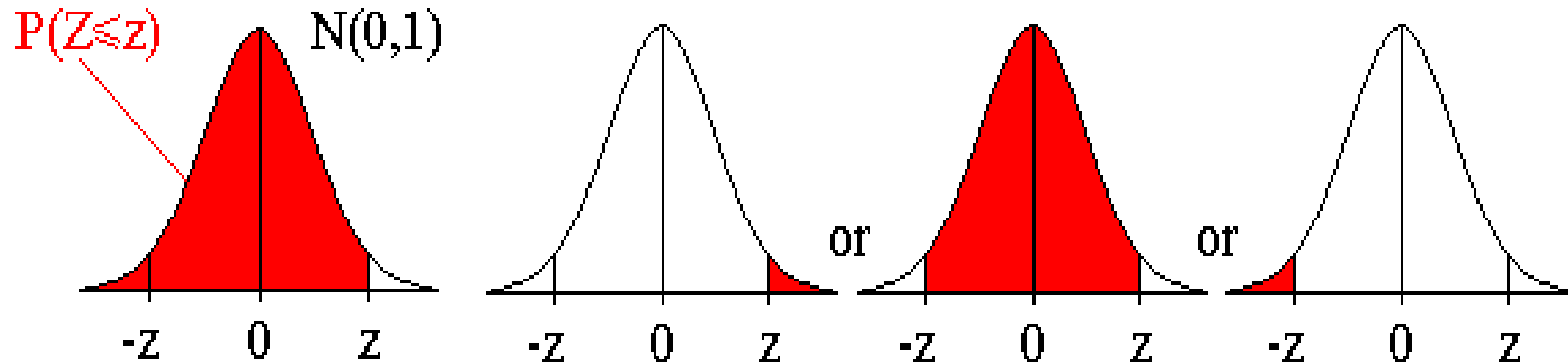


Table of Normal Distribution, $P(Z \leq z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6954	0.6993	0.7031	0.7069	0.7106	0.7143	0.7179	0.7215	0.7251
0.6	0.7291	0.7327	0.7361	0.7396	0.7430	0.7464	0.7498	0.7531	0.7564	0.7599
0.7	0.7643	0.7675	0.7708	0.7740	0.7772	0.7804	0.7836	0.7868	0.7899	0.7930
0.8	0.7969	0.7999	0.8029	0.8059	0.8088	0.8117	0.8146	0.8175	0.8203	0.8231
0.9	0.8259	0.8286	0.8312	0.8338	0.8364	0.8389	0.8415	0.8440	0.8465	0.8490
1.0	0.8524	0.8548	0.8572	0.8596	0.8619	0.8643	0.8666	0.8689	0.8712	0.8734
1.1	0.8758	0.8779	0.8801	0.8823	0.8844	0.8865	0.8886	0.8906	0.8926	0.8945
1.2	0.8965	0.8984	0.9003	0.9022	0.9041	0.9059	0.9077	0.9095	0.9112	0.9129
1.3	0.9146	0.9162	0.9179	0.9196	0.9212	0.9228	0.9244	0.9259	0.9274	0.9289
1.4	0.9306	0.9320	0.9334	0.9348	0.9361	0.9375	0.9388	0.9401	0.9414	0.9427
1.5	0.9441	0.9454	0.9467	0.9479	0.9491	0.9503	0.9515	0.9526	0.9537	0.9548
1.6	0.9559	0.9569	0.9579	0.9589	0.9599	0.9608	0.9617	0.9626	0.9635	0.9644
1.7	0.9653	0.9661	0.9669	0.9677	0.9685	0.9692	0.9699	0.9706	0.9713	0.9719
1.8	0.9726	0.9732	0.9738	0.9744	0.9750	0.9755	0.9761	0.9766	0.9771	0.9776
1.9	0.9780	0.9785	0.9790	0.9795	0.9799	0.9804	0.9808	0.9812	0.9816	0.9820
2.0	0.9824	0.9828	0.9832	0.9836	0.9840	0.9844	0.9848	0.9851	0.9854	0.9857
2.1	0.9860	0.9863	0.9866	0.9869	0.9872	0.9875	0.9878	0.9881	0.9884	0.9887
2.2	0.9889	0.9891	0.9893	0.9895	0.9897	0.9899	0.9901	0.9903	0.9905	0.9906
2.3	0.9908	0.9909	0.9910	0.9911	0.9912	0.9913	0.9914	0.9915	0.9916	0.9917
2.4	0.9918	0.9918	0.9919	0.9920	0.9921	0.9921	0.9922	0.9922	0.9923	0.9923
2.5	0.9924	0.9924	0.9925	0.9925	0.9926	0.9926	0.9926	0.9927	0.9927	0.9927
2.6	0.9928	0.9928	0.9928	0.9929	0.9929	0.9929	0.9929	0.9929	0.9930	0.9930
2.7	0.9930	0.9931	0.9931	0.9931	0.9931	0.9931	0.9931	0.9931	0.9932	0.9932
2.8	0.9932	0.9932	0.9932	0.9932	0.9932	0.9932	0.9932	0.9932	0.9932	0.9932
2.9	0.9933	0.9933	0.9933	0.9933	0.9933	0.9933	0.9933	0.9933	0.9933	0.9933
3.0	0.9934	0.9934	0.9934	0.9934	0.9934	0.9934	0.9934	0.9934	0.9934	0.9934

Eg:

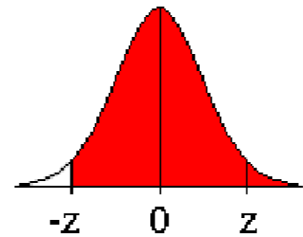
$$P(Z < 0.21) = 0.5832$$

Prob. Distribution of Continuous r.v.

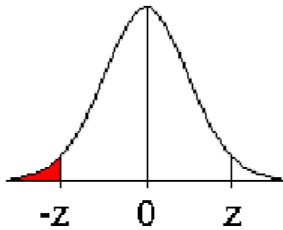
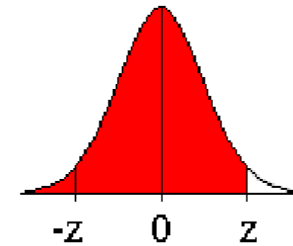
The $P(Z \leq z)$ table gives area under the curve from $-\infty$ to z , where $0 < z < 3.09$

For other values, we perform some manipulations.

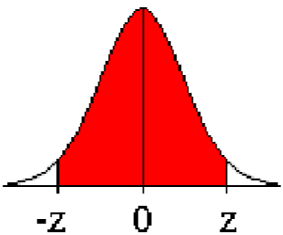
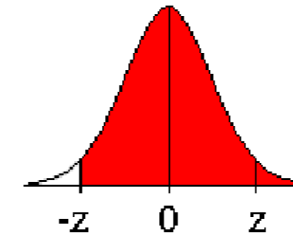
Eg:



$$P(Z \geq -z) = P(Z \leq z)$$



$$P(Z \leq -z) = 1 - P(Z \geq -z)$$



$$P(-z \leq Z \leq z) = 1 - 2P(Z \leq -z)$$

Prob. Distribution of Continuous r.v.

Probabilities for normal distributions other than the standard normal distribution $N(0, 1)$ are obtained by using the formula:

$$Z = \frac{X - \mu}{\sigma}$$

to convert from $X \sim N(\mu, \sigma^2)$ to $Z \sim N(0, 1)$ and then using the table of probabilities for the standard normal distribution $N(0, 1)$.

Prob. Distribution of Continuous r.v.

Normal distribution

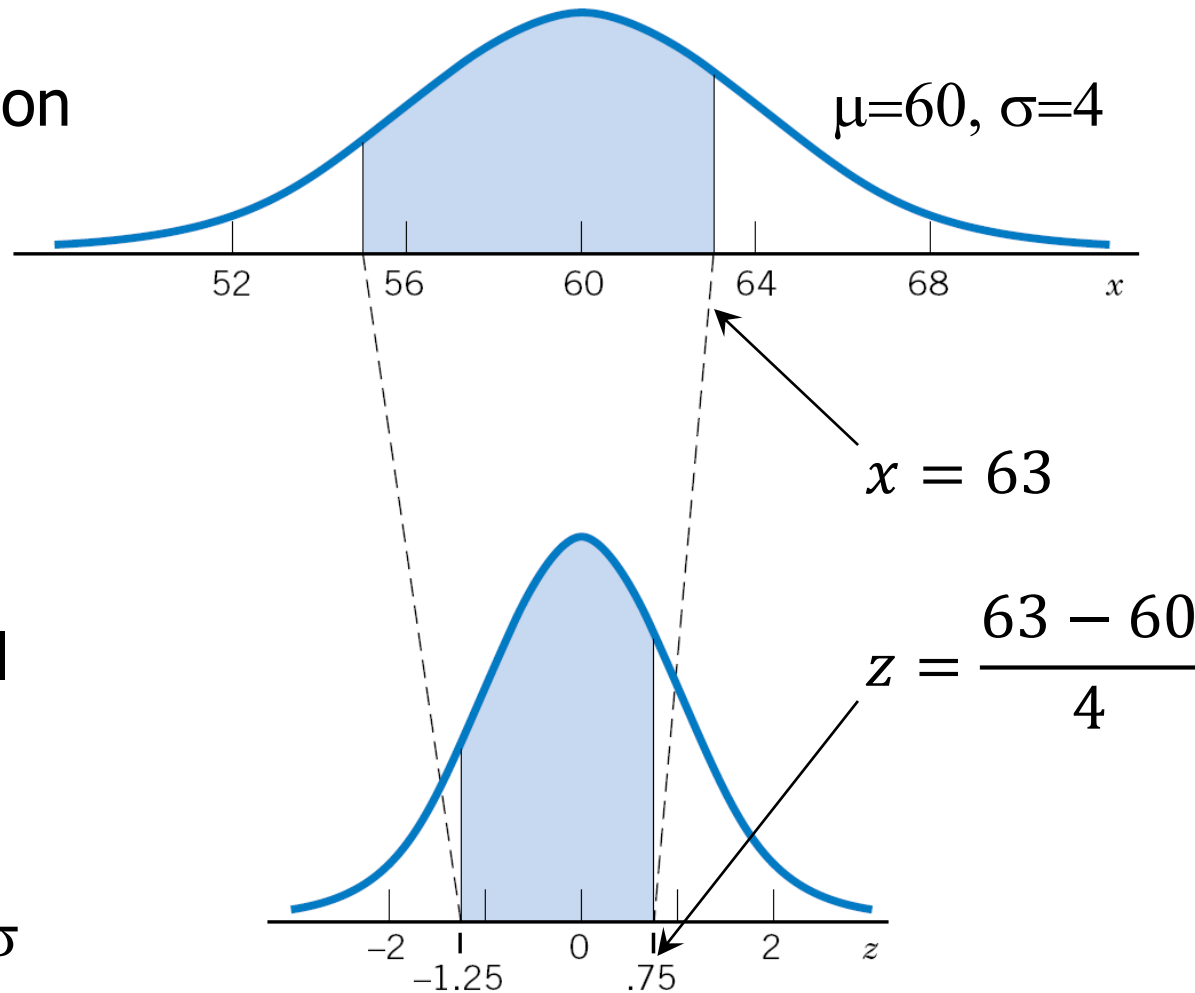
$$X \sim N(\mu, \sigma^2)$$



Standard Normal

$$Z \sim N(0, 1)$$

where $Z = (X - \mu) / \sigma$



Prob. Distribution of Continuous r.v.

Eg: A machine fills bottles with 300 ml of soft drinks. However, the actual quantity filled varies according to the normal distribution with $\mu = 298$ ml and $\sigma = 3$ ml.

What is the probability that an individual bottle contains less than 295 ml?

Prob. Distribution of Continuous r.v.

Solution

Let X denote the quantity in an individual bottle.
Given $X \sim N(298, 3^2)$, we want to find $P(X < 295)$

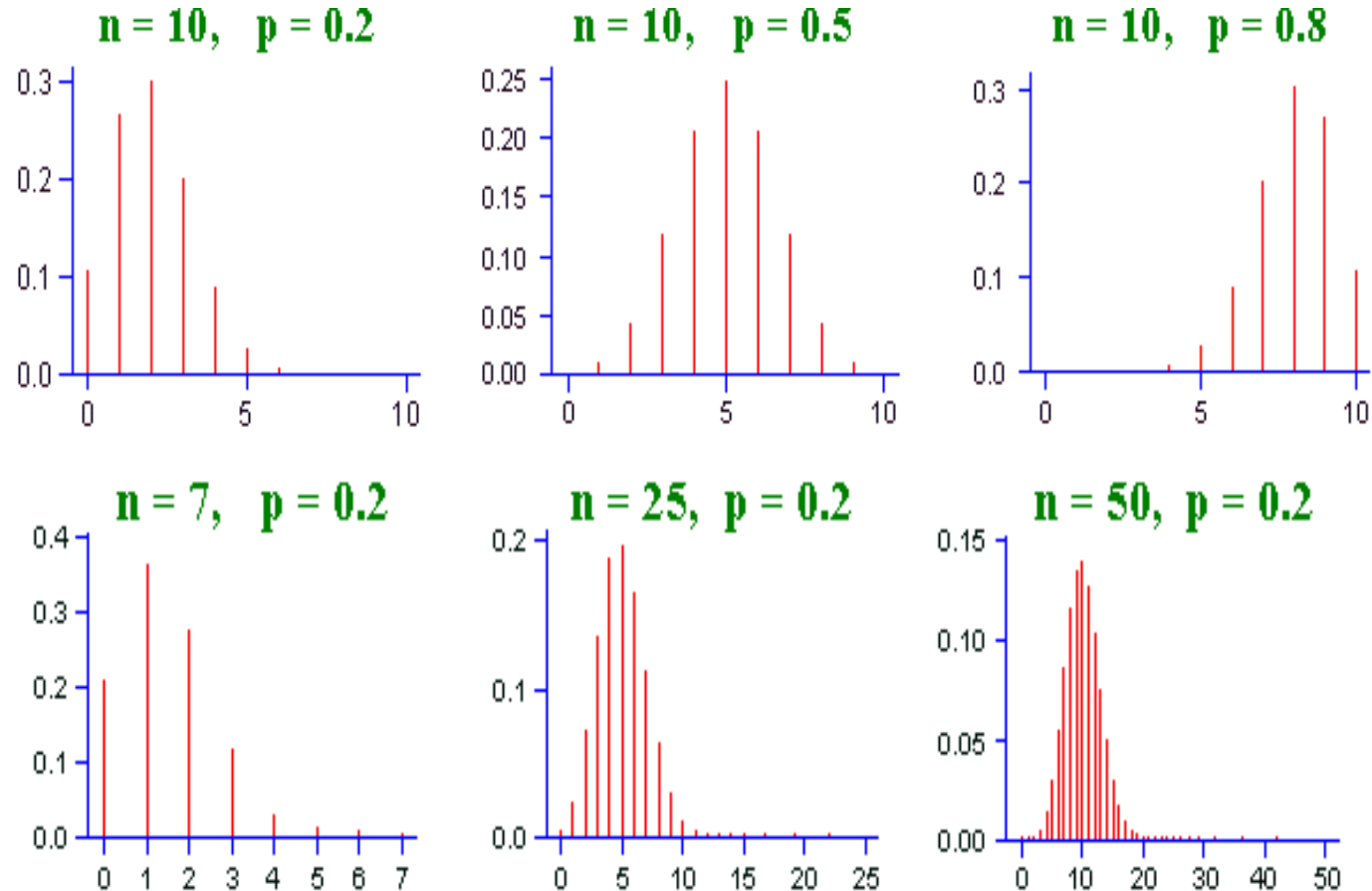
$$\text{If } X=295 \text{ then } Z = \frac{295 - 298}{3} = \frac{-3}{3} = -1$$

$$\begin{aligned} P(X < 295) &= P(Z < -1) = 1 - P(Z < 1) \\ &= 1 - 0.8413 \quad (\text{from tables}) \\ &= 0.1587 \end{aligned}$$

About 16% of bottles would have less than 295 ml.

Approx. to Binomial Distribution

For large n or p near 0.5 the binomial distribution approximately follows the normal distribution.



Approx. to Binomial Distribution

If $X \sim B(n, p)$ where n is large and p not too near 0 or 1, then X can be approximated by a normal distribution with $E(X) = np$ and $\text{var}(X) = npq$, where $q = 1 - p$.

So $Z = \frac{X - np}{\sqrt{npq}}$ is approximately $N(0, 1)$.



This approximation is reasonably good when $np > 5$ and $n(1 - p) > 5$.

Approx. to Binomial Distribution

Continuity Correction and Accuracy

For more accurate values of binomial probabilities, the approximation is improved using the continuity correction. This method considers that each whole number occupies the interval from 0.5 below to 0.5 above it.

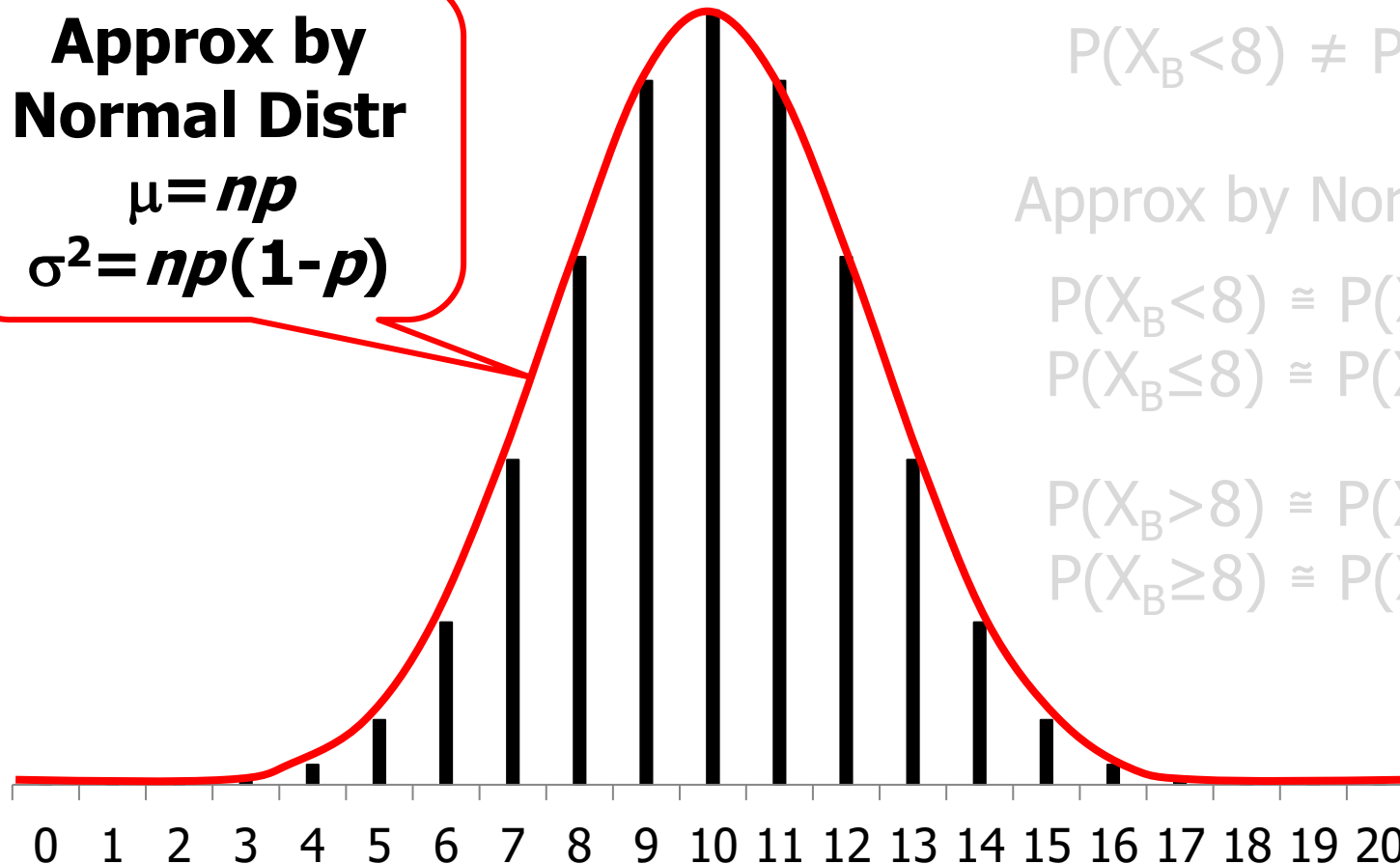
When an outcome X is to be included in the probability calculation, the normal approximation uses either $(X-0.5)$ or $(X+0.5)$.

Approx. to Binomial Distribution

Binomial Distribution $X_B \sim B(n, p)$

**Approx by
Normal Distr**

$$\mu = np$$
$$\sigma^2 = np(1-p)$$



Binomial r.v.

$$P(X_B < 8) \neq P(X_B \leq 8)$$

Approx by Normal distr

$$P(X_B < 8) \approx P(X_N < 7.5)$$

$$P(X_B \leq 8) \approx P(X_N < 8.5)$$

$$P(X_B > 8) \approx P(X_N > 8.5)$$

$$P(X_B \geq 8) \approx P(X_N > 7.5)$$

Approx. to Binomial Distribution

Eg: In a particular faculty 60% of students are men and 40% are women. In a sample of 50 students what is the probability that more than half are women?

- 1st compute the approximate probability with no continuity correction
- 2nd compare the actual result and check the accuracy of the above approximation
- 3rd compute the approximate probability with continuity correction and compare all results

Approx. to Binomial Distribution

We need to find $P(X > 25)$. Note: not $P(X \geq 25)$

No continuity correction

More than $\frac{1}{2}$
are women

$$\text{So } P(X > 25) = P\left(Z > \frac{25 - 20}{\sqrt{12}} = 1.44\right)$$

$$= 1 - P(Z \leq 1.44)$$

$$= 1 - 0.9251 = 0.075$$

The exact ans. calculated from binomial probabilities

$$\text{is: } P(X > 25) = P(X = 26) + P(X = 27) + \dots + P(X = 50)$$

$$= 0.0573$$

Approx. to Binomial Distribution

The approximate probability, using the continuity correction, is

Continuity correction

$$\begin{aligned}P(X > 25) &= P\left(Z > \frac{25.5 - 20}{\sqrt{12}}\right) \\&= P(Z > 1.59) \\&= 1 - P(Z \leq 1.59) \\&= 0.0559\end{aligned}$$

Compare to 0.0573

which is a more accurate approximation than the original approximation (without correction) of 0.075.

Approx. to Binomial Distribution

The value 25.5 was chosen as the outcome 25 was not to be included but the outcomes 26, 27, ..., 50 were to be included in the calculation.

Similarly, if the example required the probability that less than 18 students were women, the continuity correction would require the calculation

$$P(X < 18) = P\left(Z < \frac{17.5 - 20}{\sqrt{12}}\right)$$

Summary

- Random Variables
- Expected Values and Variance
- Discrete Distributions
 - 1) Binomial
 - 2) Poisson
 - 3) Geometric
- Continuous Distributions
 - 1) Uniform
 - 2) Exponential
 - 3) Normal and Standard Normal
- Normal Approximation to Binomial

Distribution	Probability	Mean	Variance
$X \sim B(n, p)$	$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$	np	$np(1 - p)$
$X \sim \text{Pois}(\mu)$	$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$	μ	μ
$X \sim G(p)$	$P(X = x) = (1 - p)^{x-1} p$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
$X \sim U(a, b)$	$f(x) = \begin{cases} \frac{1}{b - a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$	$\frac{a + b}{2}$	$\frac{(a - b)^2}{12}$
$X \sim \text{Exp}(\lambda)$	$f(x) = \lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$X \sim N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	μ	σ^2